Application of visual prompts in the domain of medical image processing BRATS

Paper09

Abstract

Transformer-based architectures, originally developed for natural language processing, have recently demonstrated significant potential in image analysis, surpassing traditional convolutional neural networks in many tasks. In this work, we investigate their application to brain tumor segmentation in medical imaging, a domain where annotated data is often limited. We propose a two-stage segmentation pipeline that incorporates visual prompting, allowing the model to leverage intermediate guidance derived from initial predictions to improve final segmentation accuracy. The approach is evaluated on publicly available brain tumor datasets, demonstrating improved performance in capturing tumor boundaries and fine structures compared to baseline methods. Our results highlight the effectiveness of combining transformers with visual prompting for medical image segmentation, particularly in low-data scenarios, and suggest a promising direction for future research in automated medical image analysis.

Keywords: Computer Vision, Medical Image Processing, Visual Prompts, Vision Transformers, BRATS

1 Introduction

Brain tumors, particularly gliomas, represent some of the most aggressive and life-threatening forms of cancer, accounting for a significant proportion of primary brain tumors in adults. Their infiltrative nature and heterogeneous growth patterns pose major challenges for precise delineation, which is crucial for treatment planning and monitoring disease progression. Magnetic resonance imaging has become the gold standard for non-invasive visualization of brain tumors, providing detailed information about tumor size, location, and morphology. However, despite advances in imaging technology, accurately defining tumor boundaries remains difficult, often requiring labor-intensive manual annotation.

Recent advances in deep learning, particularly convolutional neural networks, have enabled significant improvements in medical image analysis, providing automated tools for tumor detection and segmentation. In a study, presented by [4], the authors report, that these neural network based approaches typically achieve high Dice similarity scores, often exceeding 90% for whole tumor regions, but still struggle to provide precise boundary delineation across heterogeneous tumor subregions. More recently, Vision Transformers have emerged as a powerful alternative, leveraging self-attention mechanisms to capture long-range dependencies and global context in images. This ability to model relationships across distant image regions is particularly valuable in medical imaging, where precise delineation of heterogeneous tumors is required.

Despite their potential, the application of deep learning in medical imaging faces a critical challenge: the limited availability of high-quality annotated data. Collecting and curating expert-labeled datasets is costly and time-consuming, which can hinder the training of large models. To address this limitation, the concept of visual prompting has been proposed, where a model is guided interactively using visual cues such as points, bounding regions, or coarse masks. Prompts enable efficient adaptation of pre-trained models to small datasets.

In this paper, we propose a two-stage binary segmentation pipeline for brain tumor MRI images that combines the strengths of transformer-based architectures with visual prompting. In the first stage, an initial model generates coarse tumor masks. From these masks, representative prompts are extracted and provided as additional input to a second transformer model, which produces refined segmentations. This approach allows the model to focus on relevant tumor regions and handle cases with multiple tumor subregions, improving segmentation accuracy in low-data scenarios.

Compared to existing deep-learning visual prompting methods, we introduce in this paper the following contributions:

- *Two-stage segmentation pipeline* that combines coarse mask generation with visual prompting for final refinement.
- *Automated method to extract robust visual prompts* from preliminary masks, ensuring that the final segmentation model focuses on clinically relevant tumor regions.

2 Related Work

2.1 Promptable Medical Image Segmentation

Recent years have seen growing interest in adapting large, promptable segmentation models to medical imaging tasks. Among these, the Segment Anything Model

(SAM), proposed in [8], has emerged as a foundation model capable of general-purpose segmentation via visual prompts such as points, bounding boxes, and masks. While SAM demonstrates strong performance on natural images, its direct application to medical data remains challenging due to significant domain shifts and the absence of medical-specific pretraining.

To address these limitations, [3] introduced *SAM-Med2D*, a large-scale adaptation of SAM for 2D medical image segmentation. The authors utilized large-scale data curation - a dataset of $\sim$4.6M images and 19.7M masks across 10 modalities and 31 organs, and proposed a lightweight domain adaptation strategy. Instead of fully fine-tuning the image encoder, authors start with freezing image encoder while inserting lightweight adapter layers in each Transformer block to inject domain-specific knowledge at low cost, allowing efficient domain specialization while preserving generalization capabilities. The prompt encoder and mask decoder are updated via simulated interactive training over nine iterations, progressively refining predictions with additional points sampled from error regions.

Additionally, SAM-Med2D leverages both sparse and dense prompts during training and inference, enabling robust interactive segmentation. Experimental results demonstrate substantial improvements in Dice score over the original SAM and fine-tuned baselines across multiple anatomical regions and imaging modalities.

2.2 SAM-Based Brain Tumor Segmentation

The application of SAM, developed by [8], to brain tumor segmentation has been explicitly studied by [14], who evaluated SAM on 2D slices derived from the brain tumor segmentation dataset. Given SAM's 2D design, the authors converted volumetric MRI scans into axial slices and applied various prompt strategies, including point prompts, bounding boxes, and their combinations. Initial experiments using a frozen SAM model revealed limited performance, motivating a fine-tuning strategy where only the mask decoder parameters were updated.

Their findings indicate that bounding box prompts yield the highest segmentation accuracy, while an excessive number of point prompts can degrade performance due to over-segmentation. The results also highlight variability across MRI modalities, with superior performance on T2 and FLAIR images. Although fine-tuning significantly improved segmentation accuracy, the study underscores the sensitivity of SAM-based approaches to prompt quality and noise, particularly in realistic interactive settings.

2.3 Robust Interactive Segmentation

Beyond SAM-based approaches, [9] proposed PRISM, a promptable and robust interactive segmentation framework designed for 3D medical images. Unlike SAM-derived methods, PRISM employs a hybrid encoder combining convolutional and transformer-based representations to capture both local and global image context. The model supports a wide range of prompt types, including points, bounding boxes, scribbles, and masks, and introduces an iterative correction mechanism that refines segmentations based on user feedback.

A key contribution of PRISM is its confidence-aware learning strategy, which produces multiple segmentation hypotheses alongside confidence scores, enabling reliable output selection during inference. Extensive experiments across multiple tumor segmentation benchmarks demonstrate that PRISM achieves near-human performance in interactive settings, particularly when combining multiple prompt types over successive iterations.

2.4 Metrics

When assessing segmentation models in medical imaging, it is crucial to determine how closely the predicted mask matches the ground-truth annotation. In this paper, inspired by [12], we employ one of the most commonly used metric for evaluating spatial overlap accuracy: the *Dice Coefficient*. Throughout the paper, the Dice metric will be referenced frequently, as it has become a standard evaluation metric in the medical imaging community and is directly comparable with results reported in related work. It provides robust insight into how well a model delineates tumor boundaries, and is particularly suitable for imbalanced datasets, which are common in medical tasks.

The Dice coefficient measures the similarity between the predicted mask P and the ground-truth mask G based on their overlap. Its value ranges from 0 to 1, where 1 indicates perfect agreement. The formula is:

$$Dice(P,G) = \frac{2|P \cap G|}{|P| + |G|}. \tag{1}$$

As mentioned by authors of [12], this metric is highly sensitive to boundary inaccuracies, which often occur in lesion segmentation where the target region occupies only a small portion of the image.

Using Dice as the primary metric allows us to consistently evaluate the segmentation quality across different versions of the model and to provide a fair comparison between approaches both with and without visual prompting.

3 Method

Dataset We will be using the publicly available *Brain Tumor Segmentation Challenge 2021* (BraTS) dataset from [1]. BraTS is a comprehensive benchmark designed for brain tumor segmentation and radiogenomic classification, introduced during the BraTS 2021 challenge marking its 10th anniversary. It contains pre-operative multi-parametric MRI scans from 2040 patients diagnosed with gliomas, the most common primary malignant tumors of the central nervous system. As authors of [10] describe, each 3D scan includes four MRI modalities: T1 (anatomical structure), T1ce (T1 with contrast agent highlighting active tumor regions), T2 (emphasizing edema and water content), and FLAIR (highlighting tumor-related edema by suppressing fluid signals). Expert annotations identify three tumor subregions: enhancing tumor (ET), tumor core (TC), and whole tumor (WT). All images are stored in NIfTI format as 3D volumes.

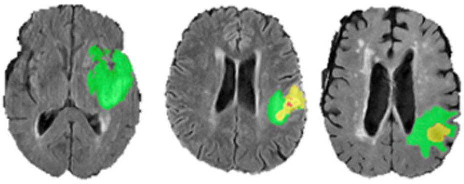

Figure 1: Example axial slice from BraTS 2021 dataset with annotated tumor regions: ET (red), TC (green), and WT (blue), taken from [1].

All scans undergo preprocessing to ensure consistency, including co-registration to a common anatomical template, resampling to $1 \times 1 \times 1$ mm^3 voxel size, skull stripping, and normalization to dimensions $240 \times 240 \times 155$ voxels. This standardized representation supports robust training and evaluation of segmentation models. For fine-tuning of the segmentation models, we will use 80% of the dataset for training, 10% for validation and 10% for testing.

Pipeline The proposed approach combines a ViT architecture with interactive visual prompting. The pipeline consists of two stages: a foundation ViT model generates an initial coarse tumor mask from a 2D slice of the 3D MRI volume, and a domain-adapted ViT model refines segmentation using the original slice and a point-based visual prompt derived from the initial mask. This design enhances interactive control and segmentation accuracy while leveraging large pre-trained models for domain adaptation.

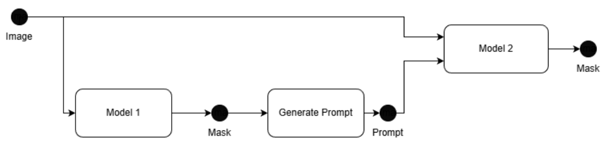

Figure 2: Overview of the proposed two-stage segmentation pipeline.

The first stage outputs a spatially consistent but coarse mask, which is converted into a point prompt representing tumor location. This prompt guides the second model to integrate local and global context for precise pixel-wise classification. The second stage produces the final segmentation mask, capturing fine tumor boundaries and complex spatial relationships.

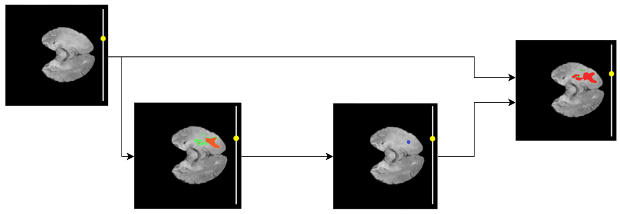

Figure 3: Visual illustration of the proposed pipeline: green denotes ground-truth annotation, orange the initial mask, blue the point prompt, and red the final refined mask.

This modular design enables separate evaluation of each stage and supports adaptation to limited-data scenarios by leveraging strong foundation models for coarse localization and domain-specific transformers for fine segmentation.

Model Selection SAM (proposed by [8]) is a foundation model for image segmentation trained on over one billion masks. It supports prompt-based segmentation using points, bounding boxes, or masks, making it highly adaptable. SAM's architecture comprises three modules: an image encoder (ViT-based), a prompt encoder, and a lightweight mask decoder. In our pipeline, SAM processes a 2D slice from the 3D MRI volume to generate an initial tumor mask and from it the point prompt.

MiT (proposed by [13]) is a hierarchical transformer designed for segmentation tasks. It combines global attention with multi-scale feature extraction, producing rich representations at different resolutions. MiT uses overlapping patch embeddings, Mix-FFN blocks with depthwise convolutions, and an efficient All-MLP decoder to aggregate multi-level features. This architecture is well-suited for medical imaging, where fine boundary details and global context are critical. In proposed pipeline, MiT receives the 2D slice from the 3D MRI volume and the point prompt as input, inspired by [2], to generate the final segmentation mask.

Both these models will be fine tuned on the BraTS 2021 dataset to learn the domain knowledge required for accurate brain tumor segmentation.

3.1 Prompt generation

After SAM produces an initial mask, we derive a point-based visual prompt for MiT to guide refinement.

The process at first involves detecting connected components in the SAM mask and selecting the largest region R to minimize artifacts (also decided to not go with multiple regions because of findings of [14] about multiple point prompts lowering model precision).

The second step involves computing a distance transform to find the pixel farthest from the region boundary:

$$D(x,y) = \min_{(i,j)\in\partial R} \sqrt{(x-i)^2 + (y-j)^2}, \quad (2)$$

where ∂R denotes the boundary of region R. The pixel with the maximum $D(x,y)$ is chosen as the point prompt.

3.2 BCE-Dice Loss

To optimize both models while fine-tuning, we use a combined loss function of Binary Cross-Entropy (BCE) and Dice loss, which is effective for imbalanced medical datasets:

$$\mathscr{L}_{total} = \lambda_{BCE} \cdot \mathscr{L}_{BCE} + \lambda_{Dice} \cdot \mathscr{L}_{Dice}. \quad (3)$$

BCE penalizes pixel-wise misclassification:

$$\mathscr{L}_{BCE} = -\frac{1}{N}\sum_{i=1}^{N} \left[y_i \log(\hat{y}_i) + (1-y_i)\log(1-\hat{y}_i) \right], \quad (4)$$

while Dice loss measures overlap between prediction and ground truth:

$$\mathscr{L}_{Dice} = 1 - \frac{2\sum_i \hat{y}_i y_i}{\sum_i \hat{y}_i + \sum_i y_i}. \quad (5)$$

This combination ensures stable gradients and improved convergence, particularly for small tumor regions.

4 Results

Experimental setup We evaluate three configurations on BraTS 2021: (1) a fine-tuned *Segment Anything Model (SAM)* [8], (2) a fine-tuned *MixVision Transformer (MiT)* [13], and (3) the proposed two-stage $SAM \rightarrow MiT$ pipeline. SAM is trained on axial 2D slices (256×256; internally upscaled to 1024×1024 to match its positional encodings) with four MRI modalities concatenated as channels; MiT consumes the same slice plus a fifth channel encoding the point prompt derived from SAM's mask. Both models are optimized for binary segmentation (tumor vs. background) using $BCE + Dice$ loss with learning rate 1×10^{-5}, up to 50 epochs, batch sizes of 128 (SAM) and 256 (MiT), and decision threshold 0.7 on the validation set. The best checkpoints are selected by validation Dice.

4.1 Quantitative Results

Fine-tuned SAM achieves Dice scores of 96.47% on training split, 95.45% validation split, and 94.82% on testing split. Fine-tuned MiT reaches Dice scores of 96.45% on training split, 96.23% on validation split, and 95.49% on testing split. The two-stage $SAM \rightarrow MiT$ pipeline attains Dice score of **95.83%** on the testing split. After visualization of Dice score distribution of the model predictions results on the test set (Figures 4 and 5), we find that there is a small percentage of images with low Dice scores.

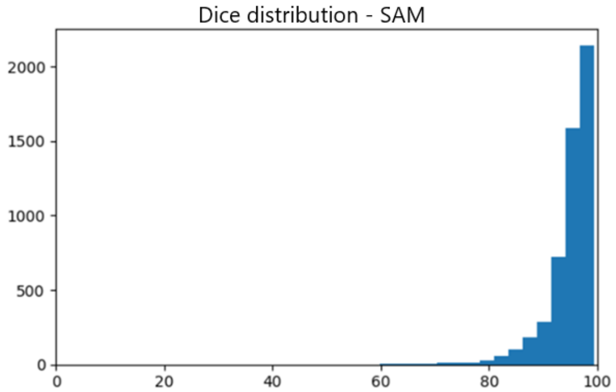

Figure 4: Distribution of Dice score across test images for fine-tuned SAM. X-axis shows Dice scores, while Y-axis represents frequency of Dice scores.

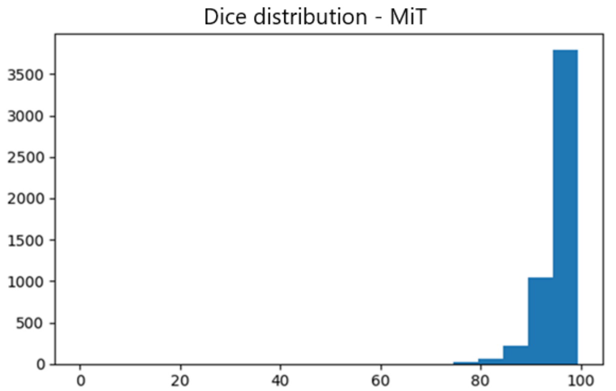

Figure 5: Distribution of Dice score across test images for fine-tuned MiT. X-axis shows Dice scores, while Y-axis represents frequency of Dice scores.

Failure and success modes Across models, low Dice outliers typically involve small, multifocal, or low-contrast lesions where boundary ambiguity is high (Figure 6).

On the other hand, success cases show large, well-delimited enhancing components (Figure 7). MiT reduces false positives near ventricular CSF and sulcal fluid compared to SAM, making it more consistent with its stronger multi-scale feature aggregation.

4.2 Comparison to Existing Solutions

To contextualize our approach, we used representative state-of-the-art (SOTA) methods and models on **binary tumor segmentation** (tumor vs. background) using the dataset we used to evaluate our approach (BraTS 2021) and report Dice scores for direct comparison. Below we summarize each approach used in the original papers.

SegResNet In [7], authors proposed a residual U-shape CNN (encoder–decoder) where residual blocks and deep supervision stabilize optimization. They used a *Hybrid*

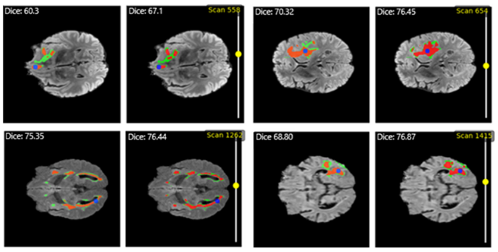

Figure 6: Worst cases of SAM and MiT visualized. Each pair shows SAM's mask and MiT's mask. On the left side is SAM's prediction and on the right side of the pair is MiT's prediction. Green denotes ground-truth annotation. Red denotes the final mask prediction. Blue denotes the point prompt.

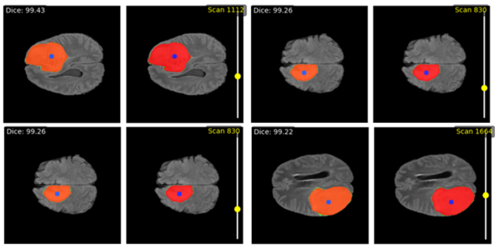

Figure 7: Best cases of SAM and MiT visualized. Each pair shows SAM's mask and MiT's mask. On the left side is SAM's prediction and on the right side of the pair is MiT's prediction. Green denotes ground-truth annotation. Red denotes the final mask prediction. Blue denotes the point prompt.

Boundary–Dice loss. The Dice loss is defined as shown in Equation 5. The Boundary loss is derived into:

$$L_B(\theta) = [F(X) \times K - Y \times K]^2 \tag{6}$$

where L_B stands for boundary loss, F is the neural network, X is the input data, K stands for Kernel, and Y is the ground truth. The final loss is addition of dice loss and boundary loss:

$$\mathrm{DICE}(\theta) + L_B(\theta) \tag{7}$$

This special loss function was shown to improve boundary adherence.

UNETR Authors from [11] present a pure ViT encoder that tokenizes patches and feeds skip connections at multiple scales to a CNN decoder (*transformer encoder & CNN decoder*). Authors originally optimized the training with combination of *Dice loss* and *Cross-Entropy loss* expressed in Equation 3.

Swin UNETR From [6], authors propose a hierarchical Swin-Transformer encoder (shifted windows for local–global attention) coupled with a CNN decoder via multi-scale skips. In training authors used *soft Dice loss* expressed as:

$$\mathscr{L}(G,Y) = 1 - \frac{2}{J}\sum_{j=1}^{J} \frac{\sum_{i=1}^{I} G_{i,j} Y_{i,j}}{\sum_{i=1}^{I} G_{i,j}^2 + \sum_{i=1}^{I} Y_{i,j}^2} \tag{8}$$

where I denotes voxels numbers, J is classes number, $Y_{i,j}$ and $G_{i,j}$ denote the probability of output and one-hot encoded ground truth for class j at voxel i, respectively.

MedNeXt A recent high-performance segmentation backbone using modernized convolutional blocks and large receptive fields presented by authors of [5]. They optimized the model with commonly used combination of *Dice loss* and *Cross-Entropy loss* expressed in Equation 3.

MedSAM Authors of [15] present a SAM-derived foundation model adapted to medical imaging with large-scale medical pretraining, supporting prompt-based segmentation. While using combination of *Dice loss* & *Cross-Entropy loss* expressed in Equation 3, they fine-tuned the model by freezing only the prompt encoder and unfreezing the image encoder and mask decoder.

Table 1: Comparison of binary brain tumor segmentation results on BraTS 2021 (tumor vs. background).

Model	Year	Dice (%)
SegResNet [7]	2021	88.78
UNETR [11]	2022	90.61
Swin UNETR [6]	2022	93.98
MedNeXt [5]	2024	94.50
MedSAM [15]	2024	94.93
Our Method	**2025**	**95.83**

Based on Table 1, our method achieves the highest accuracy among all evaluated approaches, outperforming other methods. Although the absolute differences between the top models appear relatively small, this margin is meaningful in medical image segmentation, where even minor improvements can translate into more precise tumor boundary delineation.

To further illustrate the behavior of the different models listed in Table 1, we present three qualitative comparison examples in Figures 8, 9, 10. These examples were selected to represent increasing levels of segmentation difficulty. The first case shows a simple, well-defined tumor with a regular shape, which most models can segment accurately. The second example contains a more complex structure composed of two separate tumor regions, making the task more challenging. The third example represents the most difficult scenario, featuring multiple disconnected regions and irregular shapes. This progression allows us to visually highlight how the baseline models handle tumors of varying complexity. Our method consistently outperforms these models, achieving high overlap with visually similar predictions even with raising complexity.

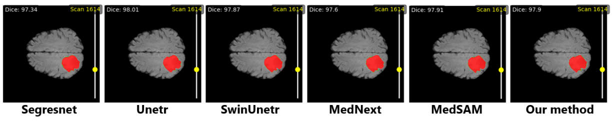

Figure 8: Qualitative comparison of tumor segmentation results across baseline models and our method on an easy-case example. All methods achieve high overlap with visually similar predictions.

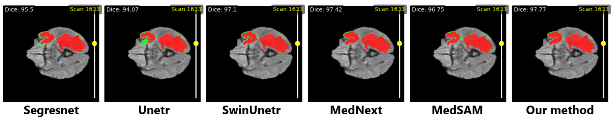

Figure 9: Qualitative comparison on a medium-complexity tumor case, containing two separate regions. Models show varying accuracy in capturing both areas, with our method providing the most accurate overlap.

5 Conclusions

This paper introduced a modular approach, where we explored prompt-guided solutions for binary tumor segmentation in BRATS domain, combining a fine-tuned foundation model for coarse localization with a fine-tuned specialized model for boundary-refined segmentation. Under a unified training protocol, the models achieved strong generalization on the public BRATS dataset. These results place our approach at or above commonly reported baselines for binary tumor segmentation.

Qualitative analyses revealed complementary strengths: SAM provides robust, spatially consistent coarse masks, whereas MiT effectively suppresses false positives and sharpens boundaries via hierarchical multi-scale features. The prompt-guided refinement consistently improved delineation of ambiguous margins without requiring manual interaction, and did so using only 2D slices and lightweight inference.

Despite these promising results, several limitations remain. First, the binary target reduces task difficulty relative to multi-class (ET/TC/WT) settings. Extending the pipeline to multi-class segmentation and assessing class-specific trade-offs is a natural next step. Second, processing slices independently limits cross-slice context. Integrating 3D volumetric cues could further stabilize predictions for small, multifocal, or low-contrast lesions.

Looking forward, we plan to generalize the pipeline to multi-class segmentation with prompt strategies tailored per subregion, explore hybrid 2D/3D architectures to incorporate volumetric context while retaining efficiency, and investigate interactive scenarios where clinicians can optionally supply additional points to correct difficult cases.

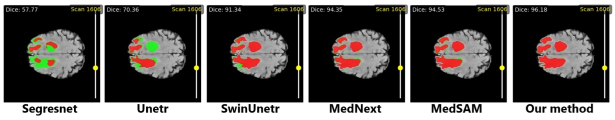

Figure 10: Qualitative comparison on a high-complexity tumor case with multiple irregular regions. Models vary widely in completeness and accuracy, while our method provides the most consistent overlap across all regions.

References

[1] Ujjwal Baid, Satyam Ghodasara, Suyash Mohan, Michel Bilello, Evan Calabrese, Errol Colak, Keyvan Farahani, Jayashree Kalpathy-Cramer, Felipe C Kitamura, Sarthak Pati, et al. The rsna-asnr-miccai brats 2021 benchmark on brain tumor segmentation and radiogenomic classification. *arXiv preprint arXiv:2107.02314*, 2021.

[2] Amir Bar, Yossi Gandelsman, Trevor Darrell, Amir Globerson, and Alexei Efros. Visual prompting via image inpainting. *Advances in Neural Information Processing Systems*, 35:25005–25017, 2022.

[3] Junlong Cheng, Jin Ye, Zhongying Deng, Jianpin Chen, Tianbin Li, Haoyu Wang, Yanzhou Su, Ziyan Huang, Jilong Chen, Lei Jiang, et al. Sam-med2d. *arXiv preprint arXiv:2308.16184*, 2023.

[4] Cecilia Diana-Albelda, Álvaro García-Martín, and Jesus Bescos. A review on deep learning methods for glioma segmentation, limitations, and future perspectives. *Journal of Imaging*, 11(8):269, 2025.

[5] Sarim Hashmi, Juan Lugo, Abdelrahman Elsayed, Dinesh Saggurthi, Mohammed Elseiagy, Alikhan Nurkamal, Jaskaran Walia, Fadillah Adamsyah Maani, and Mohammad Yaqub. Optimizing brain tumor segmentation with mednext: Brats 2024 ssa and pediatrics. *arXiv preprint arXiv:2411.15872*, 2024.

[6] Ali Hatamizadeh, Vishwesh Nath, Yucheng Tang, Dong Yang, Holger R Roth, and Daguang Xu. Swin unetr: Swin transformers for semantic segmentation of brain tumors in mri images. In *International MICCAI brainlesion workshop*, pages 272–284. Springer, 2022.

[7] Cheyu Hsu, Chunhao Chang, Tom Weiwu Chen, Hsinhan Tsai, Shihchieh Ma, and Weichung Wang. Brain tumor segmentation (brats) challenge short paper: Improving three-dimensional brain tumor segmentation using segresnet and hybrid boundary-dice loss. In *International MICCAI Brainlesion Workshop*, pages 334–344. Springer, 2021.

[8] Alexander Kirillov, Eric Mintun, Nikhila Ravi, Hanzi Mao, Chloe Rolland, Laura Gustafson, Tete

Xiao, Spencer Whitehead, Alexander C Berg, Wan-Yen Lo, et al. Segment anything. In *Proceedings of the IEEE/CVF international conference on computer vision*, pages 4015–4026, 2023.

[9] Hao Li, Han Liu, Dewei Hu, Jiacheng Wang, and Ipek Oguz. Prism: A promptable and robust interactive segmentation model with visual prompts. In *International Conference on Medical Image Computing and Computer-Assisted Intervention*, pages 389–399. Springer, 2024.

[10] Amr Mohamed, Mahmoud Rabea, Aya Sameh, and Ehab Kamal. Brain tumor radiogenomic classification. *arXiv preprint arXiv:2401.09471*, 2024.

[11] Ramtin Mojtahedi, Mohammad Hamghalam, and Amber L Simpson. Multi-modal brain tumour segmentation using transformer with optimal patch size. In *International MICCAI Brainlesion Workshop*, pages 195–204. Springer, 2022.

[12] Abdel Aziz Taha and Allan Hanbury. Metrics for evaluating 3d medical image segmentation: analysis, selection, and tool. *BMC medical imaging*, 15(1):29, 2015.

[13] Enze Xie, Wenhai Wang, Zhiding Yu, Anima Anandkumar, Jose M Alvarez, and Ping Luo. Segformer: Simple and efficient design for semantic segmentation with transformers. *Advances in neural information processing systems*, 34:12077–12090, 2022.

[14] Peng Zhang and Yaping Wang. Segment anything model for brain tumor segmentation. *arXiv preprint arXiv:2309.08434*, 2023.

[15] Kai Zhou and Dan Yu. Segmentation of brain mri tumors by medsam with prompts generated by object detection. In *2024 5th International Seminar on Artificial Intelligence, Networking and Information Technology (AINIT)*, pages 2081–2085. IEEE, 2024.